



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

PipelineIQ: Intelligent CI/CD Pipeline Management System is a DevOps-based platform designed to automate, monitor, and optimize the software development and deployment lifecycle. The system integrates Git workflows, CI/CD pipelines, containerization, automated testing, security scanning, and deployment monitoring into a unified pipeline management solution. It enables developers to automatically build, test, scan, and deploy applications whenever changes are committed to a version-control repository.

PipelineIQ incorporates tools and practices such as Git, Docker, CI/CD automation, SonarQube, Trivy, Prometheus, and Grafana to improve code quality, security, deployment reliability, and system observability. The platform provides real-time insights into pipeline execution, build status, failures, security vulnerabilities, and application performance. By automating repetitive development and deployment tasks, PipelineIQ reduces manual effort, detects issues earlier, and supports faster and more reliable software releases.

The proposed system demonstrates key DevOps and Agile software engineering principles, including continuous integration, continuous delivery, automated quality assurance, security integration, monitoring, and iterative development. Overall, PipelineIQ provides a practical and scalable approach to managing modern software delivery pipelines while improving development efficiency, security, reliability, and deployment speed.

Signature of the Guide

HOD